



Course Description	
COURSE NUMBER and NAME	DCS 403 Data Structure & Algorithms Design
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	This course introduces students to C++ data structures and how to store data using linked lists, arrays, stacks, and queues. Students learn basic algorithm design paradigms and the divide-and-conquer approach, which are used to solve a large variety of computational problems. Additionally, students learn the advanced technique of dynamic programming to help develop optimized implementations of several algorithms.
REQUIRED TEXT	Goodrich, M. T., Tamassia, R., Mount, D.M., (2011). <i>Data Structures and Algorithms in C++</i> (2nd ed.). Wiley. ISBN 978-1-11-8136638
INSTRUCTIONAL METHOD	Online / On-Campus

Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism (Rubric)	160	16%	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
Week 1 Assignment	100	10%	1, 3, 4, 6
Week 2 Assignment	100	10%	1, 3, 4, 6
Week 3 Assignment	100	10%	1, 2, 3, 4, 6
Week 4 Assignment	100	10%	1, 2, 3, 4, 5, 6
Week 5 Assignment	100	10%	7, 8
Week 6 Assignment	100	10%	8, 9
Week 7 Assignment	100	10%	8, 9
Week 8 Assignment	140	14%	10, 11, 12

Assignments	Totals	Weight	CLOs
Total Points/Percentage	1000 Points	100%	
Course Policies			

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Wednesday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Saturday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines, and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the Virtual Class Session (VCS) submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication



help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.

Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Analyze and design strategies for solving basic programming problems.	3
2. Utilize primitive data types, selection statements, loops, and methods to write programs.	3
3. Develop programs to solve a variety of problems in math, science, business, and games.	2
4. Utilize the step-wise refinement approach.	5
5. Utilize arrays to store, process, and sort data.	5
6. Utilize an IDE to develop programs.	5
7. Design recursive solutions.	3
8. Analyze algorithm complexities.	5
9. Describe and analyze sorting algorithms.	5
10. Utilize C++ STL to develop applications.	5
11. Implement classic data structures: array lists, linked lists, stacks, queues, heaps, binary trees, and hash tables.	3
12. Represent and solve problems using graph algorithms.	5



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading:
 - Chapter 1: A C++ Primer

Week 1 In-Class Activity

A compiler is a software program that translates code (e.g., C++ source code) into a machine language program. Learning about different types of compilers and understanding how to use them is a worthwhile endeavor. This will come in handy when applying your knowledge of C++ programming in coding data structures and algorithms to solve complex problems.

In this class you will be doing a lot of programming. There are many free C++ compilers that you can use to practice your code. These are couple recommendations:

- [Visual Studio Community Edition](#)
- [Dev-C++](#)
- [Xcode](#) (for Mac)

For this activity, you will select and install one of these applications onto your computer and set it up to run C++ programs.

Week 1 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. Why do computers use zeros and ones?
2. A block is enclosed inside _____.
3. _____ is the brain of a computer.
4. The extension name of a C++ source code file is _____.
5. To improve readability and maintainability, you should declare _____ instead of using literal values such as 3.14159.
6. To assign a value 1 to variable x, you write _____.
7. The ASCII of 'a' is 97. What is the ASCII for 'c'?
8. What expression would you write that will print: smith\exam1\test.txt?
9. A character is stored in _____.
10. Suppose x is 1. What is x after x -= 1?
11. Programming style is important, because _____.



12. What expression would you write to add a value 1 to variable x?
13. Write an expression that will yield the value 0.5.
14. Note that the ASCII for character A is 65. The expression 'A' + 1 evaluates to _____.
15. Write a program that prompts the user to enter two points (x1, y1) and (x2, y2) and displays their distances. The formula for computing the distance is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Note you can use pow(a, 0.5) to compute $\sqrt{a}$.

Week 2

Assignments to complete this week:

- Reading:
 - Chapter 2: Object-Oriented Design

Week 2 In-Class Activity

Computers do not understand human languages, so programs must be written in a language a computer can use. While it is true that there are hundreds of programming languages, this class is about learning to code using C++. Therefore, it's important to begin with the basics of programming in terms of selections and mathematical functions, characters, and strings using C++.

W3schools provides free resources on programming in C++ and other programming languages. As a class, go to the [C++ HOME](#) page and click on the Start [learning C++ now](#) button. Work through the available modules in the C++ TUTORIAL to gain more hands-on experience writing C++ code.

Week 2 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. Write an expression that will display the area of a circle if the radius is positive.
2. The "less than or equal to" comparison operator is _____.
3. What justifies the outputs to the left?
4. Suppose x=10 and y=10 what is x after evaluating the expression (y >= 10) || (x++ > 10).
5. Write a program that prompts the user to enter the year and month in this order and displays the number of days in the month. For example, if the user entered month 2 and year 2000, the program should display that February 2000 has 29 days. If the user entered month 3 and year 2005, the program should display that March 2005 has 31 days.
6. Write a program to sum the following series: $\frac{1}{3} + \frac{3}{5} + \frac{5}{7} + \frac{7}{9} + \frac{9}{11} + \frac{11}{13} + \dots + \frac{95}{97} + \frac{97}{99}$
7. Write a program that prompts the user to enter the number of students and each student's score, then finally displays the highest score.



Week 3

Assignments to complete this week:

- Reading:
 - Chapter 4: Analysis Tools

Week 3 In-Class Activity

C++ provides many power constructs such as loops and functions to aid in coding applications to solve complex problems. Loops control how many times an operation or a sequence of operations is performed in succession. A function is a collection of statements grouped together to perform a specific task. Because you will be incorporating these concepts in building data structures and defining algorithms, it's important to familiarize yourself with how to use them.

W3schools provides free resources on programming in C++ and other programming languages. As a class, go to the [C++ HOME](#) page and click on [C++ Functions](#), located in the menus on the left-side of the page. Work through the available modules in the C++ Functions to gain more hands-on experience writing C++ code.

Week 3 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. The signature of a function consists of what?
2. The signature of the main function in C++ is what?
3. What function expression would you use to return an uppercase letter from char variable ch?
4. What is Math.floor(3.6)?
5. What is the printout of the call nPrint('a', 4)?
6. Write a function that converts an uppercase letter to lowercase. Use the following function header: char toLowerCase(char ch)
7. Write a function to compute the following series:
$$m(i) = \frac{1}{2} + \frac{2}{3} + \dots + \frac{i}{i+1}$$
8. Write a function that computes future investment value at a given interest rate for a specified number of years.



Week 4

Assignments to complete this week:

- Reading:
 - Chapter 3: Arrays, Links, Lists and Recursion
 - Chapter 6: Lists and Iterators ADTs

Week 4 In-Class Activity

You will have to store large numbers of values during the execution of a program. C++ provides an efficient and organized approach to accomplish this task. The array is a data structure that stores a fixed-size sequential collection of elements of the same type. There are single, two-dimensional, and multidimensional arrays. This week, you will learn about arrays and their applications in programming.

Watch the following video on multi-dimensional arrays called [C++ Programming Tutorial 65 - Multidimensional Arrays and Nested Vectors](#) and discuss as a class.

Week 4 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. If you declare an array `double list[] = {3.4, 2.0, 3.5, 5.5}`, `list[1]` is _____.
2. When you pass an array to a function, the function receives _____.
3. Write a function that finds the smallest element in an array of integers using the following header: `double min(double array[], int size)`
4. Write a test program that prompts the user to enter ten numbers, invokes the function in #3, and displays the minimum value.



Week 5

Assignments to complete this week:

- Reading:
 - o Chapter 12: Stings and Dynamic Programming

Week 5 In-Class Activity

Developing a mathematical process for solving a problem is called algorithm design. To predict the performance of an algorithm is called algorithm analysis. In Chapter 12 of the textbook, you are introduced to several different algorithms. In small groups, select one of these algorithms and discuss how it works. Return and share with the class what you discovered and learned.

Week 5 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. Fill in the code to complete the following function for checking whether a string is a palindrome.

```
bool isPalindrome(const char * const s)
{
    if (strlen(s) <= 1) // Base case
        return true;
    else if _____ // Base case
        return false;
    else
        return isPalindrome(substring(s, 1, strlen(s) - 2));
}
```

2. Fill in the code to complete the following function for computing factorial.

```
/** Return the factorial for a specified index */
long factorial(int n)
{
    if (n == 0) // Base case
        return 1;
    else
        return _____; // Recursive call
}
```

3. How many times the recursive moveDisks function is invoked for 3 disks?
4. What is the number of iterations in the following loop:



```
int count = 5;  
while (count < n) {  
    count = count + 3;  
}
```

5. $O(1)$ is _____.
6. On an average, linear search searches _____.
7. Why is the analysis often for the worst case?



Week 6

Assignments to complete this week:

- Reading:
 - Chapter 5: Stacks, Queues, and Deques
 - Chapter 8: Sorting
 - Chapter 11: Sorting, Sets and Selection

Week 6 In-Class Activity

In Chapter 5 of the textbook, you are introduced to a new data structured class called *linked list*. A linked list is efficient for storing and managing elements. In small groups, discuss one of the functions in the LinkedList class and how it is implemented. Return and share with the class what you discovered and learned.

Week 6 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. Suppose you choose the first element as a pivot in the list {5 2 9 3 8 4 0 1 6 7}. Using the partition algorithm in the book, what is the new list after the partition?
2. The average-time complexity for merge sort is _____
3. The time to merge two sorted lists of size n is _____
4. The average-time complexity for quick sort is _____
5. The worst-time complexity for heap sort is _____
6. If a linked list has only one node, is head == tail true?
7. The gcd(m, n) can also be defined recursively as follows:
 - If $m \% n$ is 0, gcd (m, n) is n.
 - Otherwise, gcd(m, n) is gcd(n, m % n).

Write a recursive function to find the GCD and then write a test program that computes gcd(24, 16) and gcd(255, 25).



Week 7

Assignments to complete this week:

- Reading:
 - Chapter 7: Trees
 - Chapter 10 Search Trees

Week 7 In-Class Activity

[EDUCBA](#) provides free software development resources to help learners develop their skills in various programming languages. For this in-class activity, break into small groups and work through the tutorial on [Binary Search Trees](#). Once complete, return and discuss as a class.

Week 7 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. Visiting the current node first, then the left subtree of the current node, and finally the right subtree of the current node is called _____.
2. Visiting the left subtree of the current node first, then the current node itself, and finally the right subtree of the current node is called _____.
3. What is the process of visiting each node in the tree exactly once called?
4. A data structure to support searching, sorting, inserting, and deleting data efficiently is called a _____.
5. When a new TreeNode is created, what is the value for its left and right?
6. Write the code that displays the root's left child's left child's element.
7. Implement the search(element) function using recursion.
8. What is the time complexity of inserting an element into a BST?
9. What is the time complexity of deleting an element from a BST?
10. If a set of elements is inserted into a BST in two different orders, will the two corresponding BSTs look the same? Will the inorder traversal be the same? Will the postorder traversal be the same? Will the preorder traversal be the same?
11. The assignment operator should be implemented in the BST class along with the copy constructor and destructor. Implement the assignment operator.



Week 8

Assignments to complete this week:

- Reading:
 - Chapter 13: Graph Algorithms
 - Chapter 14: Memory Management and B-Trees

Week 8 In-Class Activity

If you're interested in a career as a programmer, you will likely have the chance to show what you know during the interview process. [EDUCBA](#) has provided a good introduction to algorithm interview questions with answers found [here](#). Break into small groups and work through the questions, examples, and answers. Once complete, return and discuss as a class.

Week 8 Assignment ([Rubric](#))

Answer the following questions related to your study:

1. What are the three main components of the STL? What are the relationships among them?
2. What are the three types of containers? What are they used for?
3. Does C++ define a base class for all containers?
4. What is a first-class container?
5. Which containers support iterators for traversing the elements in the containers?
6. Does an iterator act like a pointer to an element? How do you obtain the iterator for the first element in a container? How do you obtain the iterator that points to the position after the last element in a container?
7. List the types of iterators.
8. For what applications should you use a vector, a deque, or a list? What types of the iterators are supported in vector, deque, and list?
9. How do you remove elements in a vector, a deque, or a list?
10. Are the sort, splice, merge, and reverse functions contained in vector, deque, or list?
11. For what applications should you use a set or multiset? What are the differences between set and multiset?
12. For what applications should you use a map or multimap? What are the differences between map and multimap?
13. What is the header file for set? What is the header file for multiset? What is the header file for map? What is the header file for multimap?
14. Why container adapters are called adapters? Do container adapters have iterators?
15. Can you create a stack, queue, or priority_queue using a vector, deque, or list?
16. How do you insert elements to a priority_queue? How do you remove elements from a priority_queue? How do you find the size of a priority_queue?
17. Create an array of double values with five numbers: 1.3, 2.4, 4.5, 6.7, 9.0. Use the fill function to fill the first three elements with 5.5. Use the fill_n function to fill the first four elements with 6.9.



18. Create a deque with five numbers: 1.3, 2.4, 4.5, 6.7, 9.0. Use the generate function to fill random numbers in the deque. Use the generate_n function to fill random numbers in the deque.